



Classifying & Representing Data

Topic Test 1 Solution

Total Marks: 30

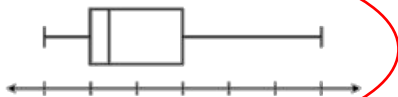
Time: 50 minutes

1 Which data collected about a virus could be classified as “categorical nominal”? 1

- A. The number of people who contracted the virus
- B. The place of origin of the virus**
- C. The rate at which the virus spread throughout the population
- D. The length of time taken to control the spread of the virus

2 Which of the data sets below represents a distribution that is positively skewed? 1

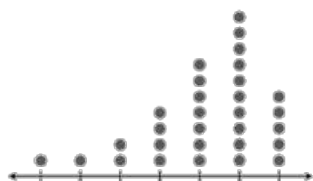
A.



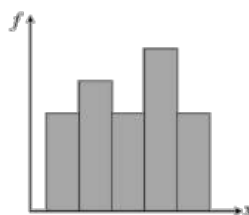
B.

stem	leaf
14	6 9
15	2 3 4 8 8
16	4 4 5 7 7 7
17	1 2 6 8 9
18	0 1

C.



D.



3 A country has 30% of the population between the ages of 20 and 30. How many people aged between 20 and 30 should be included in a sample of 250 people? $0.3 \times 250 = 75$ 1

4 A high school has 130 students in each year group from Year 7 to Year 12. A survey is to be conducted to determine the average number of hours studied per week by students at the school. Which of the following would provide the most representative sample for the survey? 1

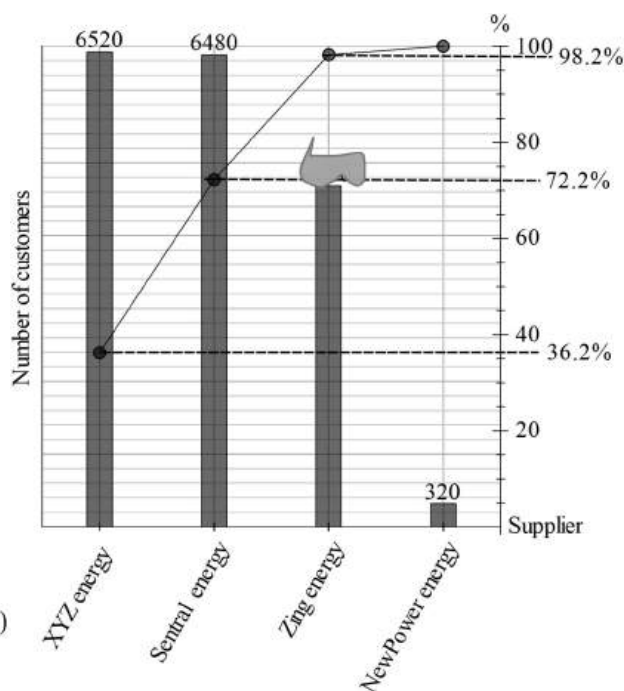
- A. Random sample
- B. Stratified sample**
- C. Systematic sample
- D. Census

5 A school's gardening program is run by 25 students, 5 teachers and 8 parents. A stratified sample of 8 people is selected to attend an environmental conference. How many teachers will attend the conference? 1

$$\frac{5}{25 + 5 + 8} \times 8 = 1$$

- 6 The Pareto chart below shows data indicating the number of customers using various energy supply companies. The number at the top of the Zing energy column has become smudged. Use the information in the graph to work out which of the following values it is closest to.

- A. 4680 B. 4860
C. 5000 D. 5240



1

Find the total number of customers (100% on the graph)

Using XYZ Energy, we get $36.2\% = 6520$

$$1\% = 180.1104972$$

$$100\% \approx 18011$$

Using Sentral Energy we get $72.2\% - 36.2\% = 36\%$

$$36\% = 6480$$

$$1\% = 180$$

$$100\% = 18000$$

Zing Energy is $98.2\% - 72.2\% = 26\%$ of total.

Find 26% of the total customers.

$$0.26 \times 18011 = 4682.86 \approx 4683$$

$$0.26 \times 18000 = 4680$$

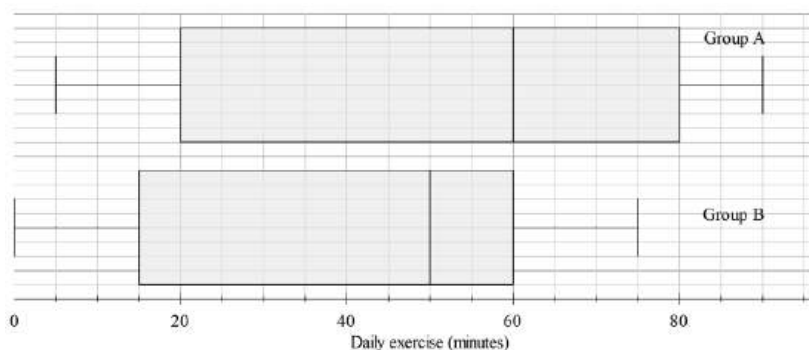
Using XYZ, our answer is closest to A.

Using Sentral, our answer is exactly A.

- 7 Employees in a workplace were divided into two groups for a fitness challenge over a week.

2

Each employee tracked the number of minutes of exercise they completed each day and reported their daily average.



The parallel box-plots above show the data collected for the two groups. There were 72

employees in Group A. The same number of employees in each group recorded an average daily exercise time of more than 60 minutes. How many employees in Group B recorded less than 15

minutes of average daily exercise time? 50% of Group A reported more than 60 minutes.
50% of 72 is 36

So 36 employees recorded more than 60 minutes in group A.

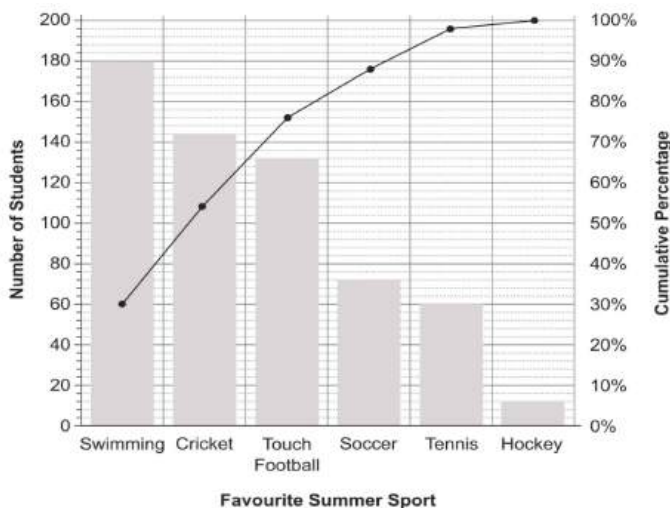
For Group B, 60 minutes corresponds to the 3rd quartile so 25% of employees in Group B scored more than 60 min.

Therefore, 25% of Group B is 36.

15 minutes daily exercise for Group B is the 1st quartile, so it also includes 25% of employees, which is 36.

- 8 A high school conducted a survey in which they asked their students "What was their favourite sport". The pareto chart below shows the data collected. What percentage of students chose soccer as their favourite sport?

12%

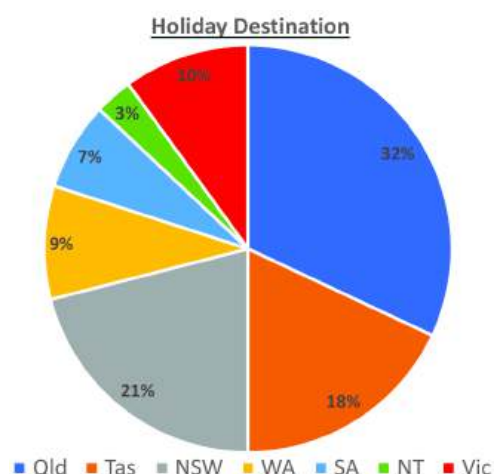


- 9 A group of people were surveyed about their favourite Australian state or territory to visit. The results are shown in the sector graph below.

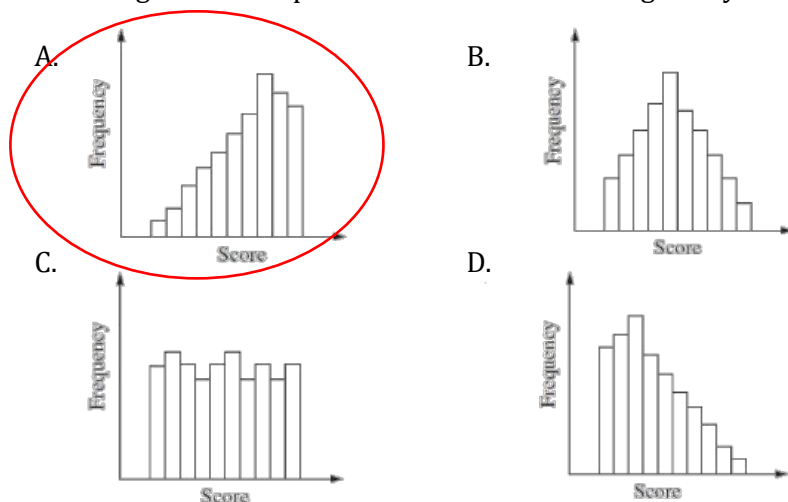
If 2475 tourists chose WA, what was the total number of people who participated in the survey?

$$0.09n = 2475$$

$$n = 27500 \text{ people}$$



- 10 Which histogram best represents a dataset that is negatively skewed?



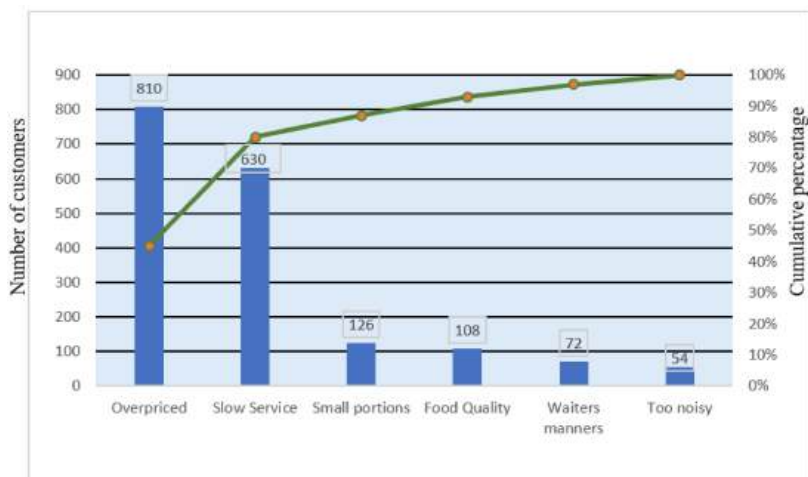
- 11 Michaela is a personal trainer. She records her client's weight as part of their program. How can the data recorded be classified?

- A. Discrete, numerical
- B. Continuous, numerical
- C. Nominal, categorical
- D. Ordinal, categorical

- 12 The military in Australia is organised in ranks such as Private or Sergeant. What sort of data is represented by these?

Ordinal

- 13 A restaurant owner collected data related to the reasons given by customers for being unhappy with his restaurant. The Pareto chart shows the data collected. Which percentage of customers



were unhappy because of the small portions?

$$\frac{126}{810 + 630 + 126 + 108 + 72 + 54} \times 100 \div 7\%$$

- 14 Stanley wants to test the wind speed in different weather conditions in his backyard. What would be the dependent variable in his experiment?

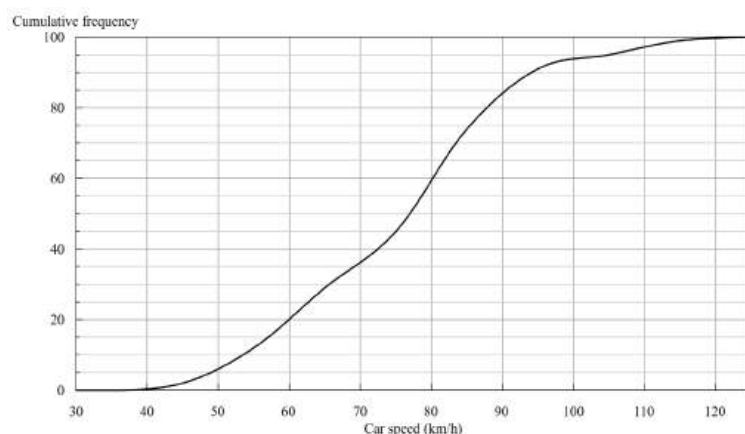
A. Location

B. Wind speed

C. Equipment used

D. Weather conditions

- 15 The cumulative frequency polygon shown below is a summary of the recorded speeds of 100 cars travelling around a bend on a country road. Around this bend, the recommended speed is 60 km/h and the actual speed limit is 80 km/h.



- a. How many of the cars were travelling above the recommended speed?

80 cars were above the recommended speed of 60 km/h

- b. The technician recording the speed of cars made the following comment:

“At least half the cars were above the speed limit.”

Is the technician’s comment correct? Justify your answer with appropriate calculations and mathematical reasoning.

The statement is incorrect. From the graph, the (approximate) median speed is 77 km/h, which is below the speed limit of 80 km/h. Half of the cars are below 77 km/h.

Alternatively – from the graph 40% of cars are above 80 km/h

16 A gym gathered information from their members on their height and weight. Which of the following is true? 1

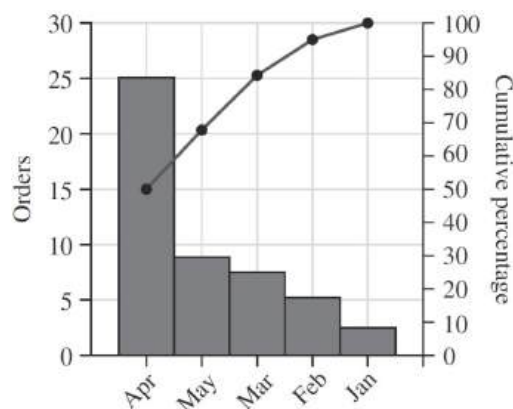
- A. Only weight is discrete data
- B. Only height is continuous data
- C. Both height and weight are discrete data
- D. Both height and weight are continuous data**

17 John works in a cake shop, and based on sales over two weeks, conducted a survey of the five most popular cakes. What type of data is this? 1

- A. Categorical nominal
- B. Categorical ordinal**
- C. Quantitative continuous
- D. Quantitative discrete

18 The Pareto chart below shows the order received by a business for five months. What percentage of orders were received in May? 1

- A. 69%
- B. 45%
- C. 30%
- D. 18%**



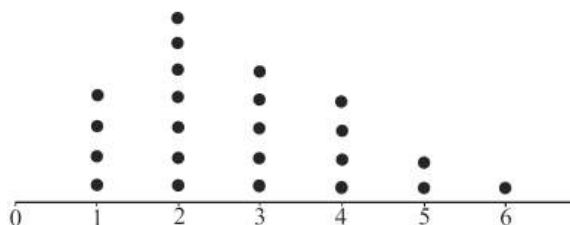
19 The number of pupils in each form group is shown in the table below. 1

Form I	Form II	Form III	Form IV	Form V	Form VI
179	183	172	176	184	181

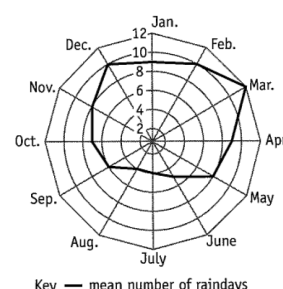
A sample of 75 pupils were surveyed using the stratified sampling method. How many of those surveyed were from Form III? 12

20 Which of the following best describes the dot plot below? 1

- A. Bimodal
- B. Symmetrical
- C. Negatively skewed
- D. Positively skewed**

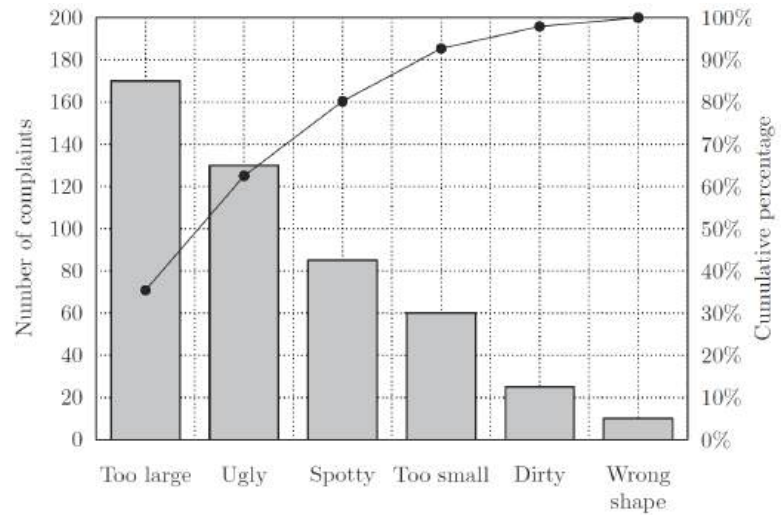


21 The graph shows the mean number of days with rainfall in a town throughout the year. What is the mean number of wet days in September and the mean number of dry days in March? a 6 b 19



22 The number of complaints for the return of a particular item is shown in the Pareto chart below. What percentage of the total number of complaints do the three most common complaints account for?

- A. 60% B. 70%
C. 80% D. 90%



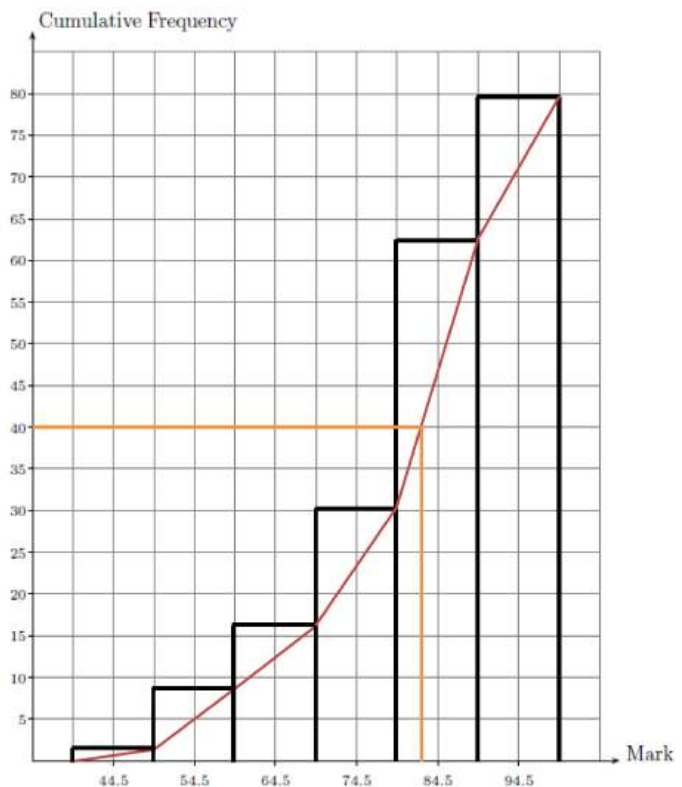
23 The results of 80 students who sat a Mathematics Standard 2 examination are recorded in the frequency table below.

- a. By drawing a cumulative frequency histogram on the grid provided below, estimate the median mark.

Median Mark : 82

- b. Calculate an estimate for the mean mark, correct to one decimal place.

Mean Mark : 79.4 (correct to one decimal place)
 Found by inputting the data using the class centres as x in 1-VAR mode with frequency turned on in the calculator.





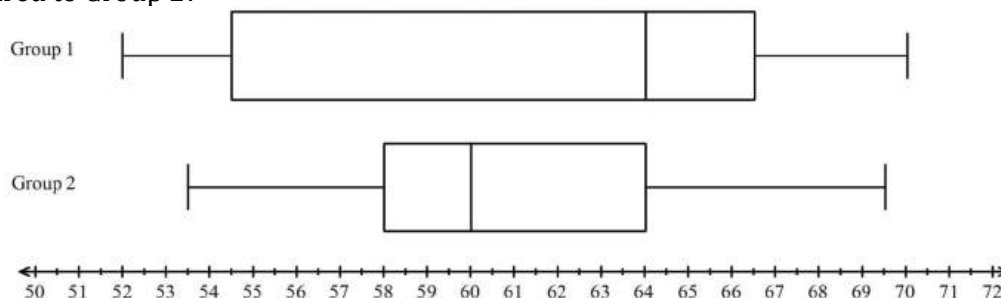
Classifying & Representing Data

Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 Two groups, with 36 students in each, have their resting heart rate recorded. The results are shown on the box-plots below. How many more students had a heart rate above 64 in Group 1 compared to Group 2?



In Group 1 64 is the median so 50% are above, or 18 students

In Group 2 64 is the upper quartile so 25% are above, or 9 students

There are 9 more students in Group 1

- 2 Barbara conducts a survey on the driving experience of 17 – 21 year-olds. She begins to design the questionnaire below to collect her data.

- a. Barbara posts the survey on her Facebook page and invites people to complete the survey. Outline why this is not a reliable way to obtain a sample of participants for the survey?

1. Do you have a drivers licence?
Yes ☐ No ☐

2. If yes, how many times did you need to take your test to get your licence?
.....

3. What size car do you drive?
Micro ☐ Small ☐ Medium ☐ Large ☐

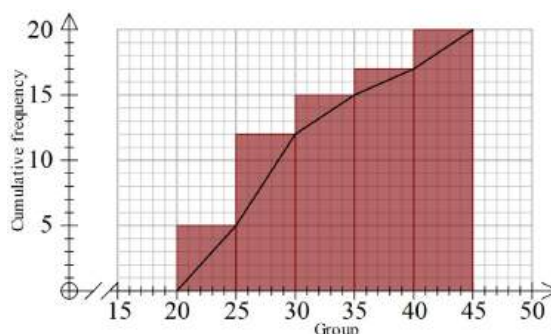
4. How far would you drive in a week?
.....

The method will not provide a random sample and is likely to be biased based on the common background and interests of Barbara's friends.

- b. Use two statistical terms to describe the type of data collected in question 3.

The data is categorical and ordinal.

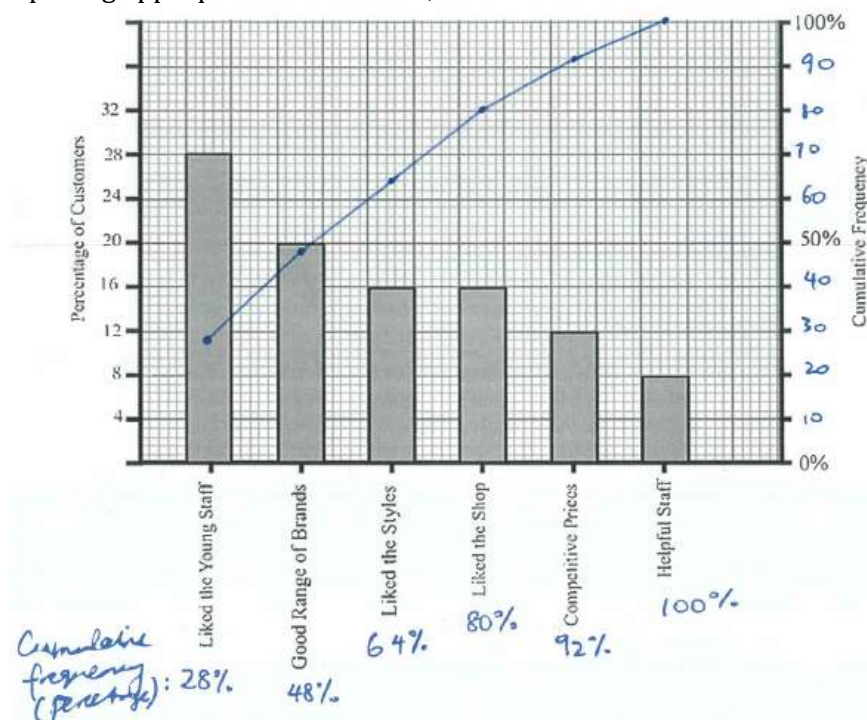
- 3 This is a cumulative frequency polygon. What is the interquartile range?



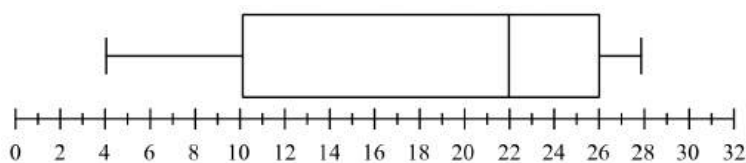
10

- 4 The chart below shows the reasons that 25 customers gave for shopping at a local clothing store. After completing appropriate calculations, draw the Pareto line on the chart above.

2



- 5 The box plot below shows the distribution of class sizes (number of students) in a school which has 64 separate classes.



- a. How many classes had a size greater than 22 students?

1

50% of 64 = 32 classes

- b. How many classes had a size between 10 students and 22 students?

1

25% of 64 = 16 classes

- 6 Lachlan works in an ice-cream store and, based on sales over two weeks, conducted a survey of the five most popular flavours. What type of data is this? **Categorical nominal**

1

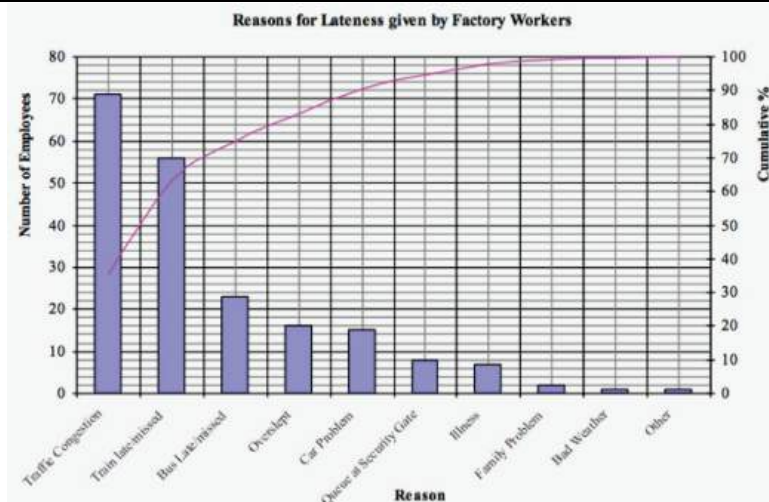
- 7 A high school has 100 students in each year group, Year 7 to Year 12. A survey is to be conducted to determine the average number of text messages sent per month by students at the school. Which of the following would provide the most representative sample for this survey?

1

- A. All Year 7 students
- B. All physics students in Year 11 and 12
- C. 20 students chosen at random from each year group**
- D. 120 students chosen at random from the school roll

The best sample would have an equal amount of people in each year randomly selected.

8 A random sample of 200 factory workers was conducted to find out the reasons why workers were late for work. The results are represented in the Pareto Chart below.



a. What percentage of

latecomers blamed traffic congestion?

$$\frac{71}{200} = 35.5\%$$

Traffic
Train late
Bus late

b. List the main causes that account for 80% of the reasons.

9 There are 2 480 phone numbers on a list. Every 8th number is phoned. What type of sampling is this?

Systematic sampling

10 Handmade chocolates are checked for size and shape. Every 30th chocolate is sampled. Which term best describes this type of sampling?

A. Census

B. Random

C. Stratified

D. Systematic

11 A survey asked the following question for students born in Australia:

"Which State or Territory were you born in?" How would the responses be classified?

A. Categorical, ordinal

B. Categorical, nominal

C. Numerical, discrete

D. Numerical, continuous

12 Ms Wiggison decided to survey a sample of 10% of the students at her school. The school enrolment is shown in the table.

Year	7	8	9	10	11	12	Total
Number of students	225	232	233	230	150	130	1200

She surveyed the same number of students in each year group. How would the numbers of students surveyed in Year 10 and Year 11 have changed if Ms Wiggison had chosen to use a stratified sample based on year groups?

A. Increased in both Year 10 and Year 11

B. Decreased in both Year 10 and Year 11

C. Increased in Year 10 and decreased in Year 11

D. Decreased in Year 10 and increased in Year 11

Total students surveyed

$$= 10\% \times 1200 = 120$$

Students surveyed per year group

$$= \frac{120}{6} = 20$$

A stratified sample would have sampled 10% of each year group.

$$\text{In Year 10, } 10\% \times 230 = 23$$

$$\text{In Year 11, } 10\% \times 150 = 15$$

$\therefore$ More students sampled in Year 10 and less in Year 11.

13 Which set of data is classified as categorical and nominal? 1

A. blue, green, yellow

B. small, medium, large

C. 5.2 cm, 6 cm, 7.21 cm

D. 4 people, 5 people, 9 people

14 A survey asked the following question. 'How many brothers do you have?' 1

How would the responses be classified?

A. Categorical, ordinal

B. Categorical, nominal

C. Numerical, discrete

D. Numerical, continuous

15 A dataset has the following five-number summary. If the range of

the dataset is 8, what is the minimum value of the dataset?

$$\text{Range} = \text{Max Value} - \text{Min Value}$$

$$8 = 15 - \text{Min Value}$$

$$\therefore \text{Min Value} = 15 - 8 = 7$$

Minimum value	?
First quartile	10
Median	12
Third quartile	13
Maximum value	15

16 The marks for a Science test and a Mathematics test are presented in box-and-whisker plots. 1

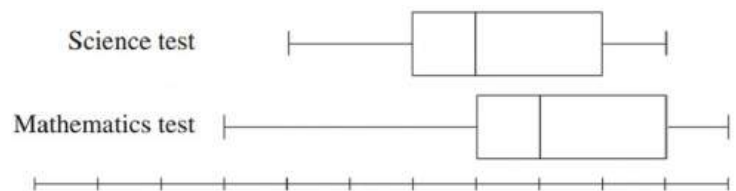
Which measure must be the same for both tests?

A. Mean

B. Range

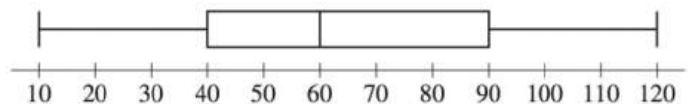
C. Median

D. Interquartile range



17 The times, in minutes, that a large group 1

of students spend on exercise per day are



presented in the box-and-whisker plot. What percentage of these students spend between 40

minutes and 60 minutes per day on exercise?

A. 17%

B. 20%

C. 25%

D. 50%

$$Q_1 = 40 \quad \text{Median} = 60$$

$$\therefore \% \text{ Students between 40 and 60}$$

$$= 50\% - 25\% = 25\%$$

18 A set of data is displayed in this box-and-whisker plot. Which of the following best describes 1

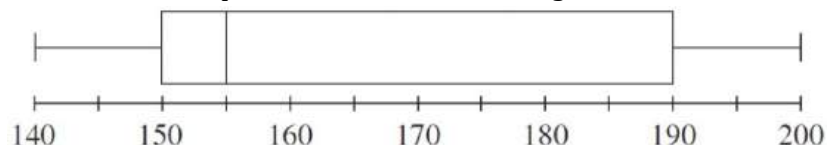
this set of data?

A. Symmetrical

B. Positively skewed

C. Negatively skewed

D. Normally distributed



- 19 The box-and-whisker plots show the results

of a History test and a Geography test. In

History, 112 students completed the test. The number of students who scored above 30

marks was the same for the History test and the Geography test. How many students completed

the Geography test?

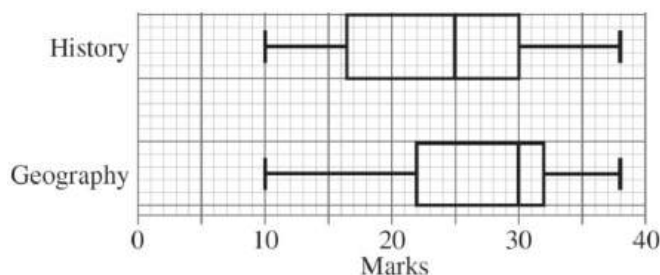
In History

$$Q_3 = 30$$

$$\therefore \text{Students scoring over 30} = 25\% \times 112 = 28$$

In Geography: Median = 30

$$\therefore \text{Students that completed Geography} = 2 \times 28 = 56$$



2

- 19 The heights of the 60 members of a choir were

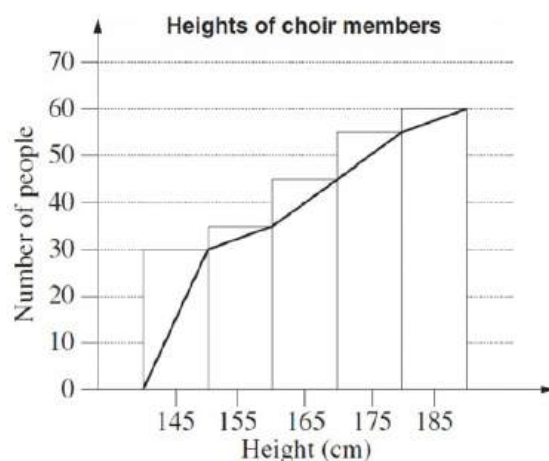
recorded. These results were grouped and then

displayed as a cumulative frequency histogram

and polygon. The shortest person in the choir is

140 cm and the tallest is 190 cm. Draw an accurate

box-and-whisker plot to represent the data.



3

$$\text{Low} = 140$$

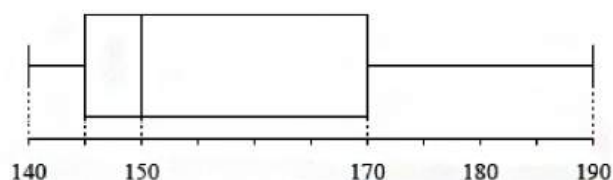
$$\text{High} = 190$$

$$\text{Median} = 150 \quad (\# \text{ People} = 30)$$

$$Q_1 = 145 \quad (\# \text{ People} = 15)$$

$$Q_3 = 170 \quad (\# \text{ People} = 45)$$

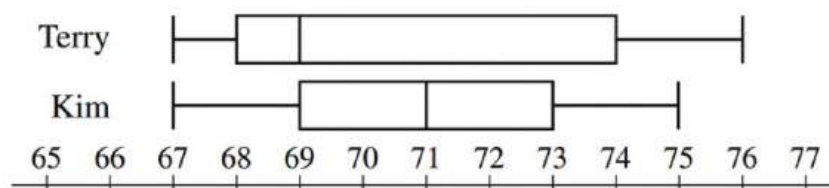
Box and Whisker



- 20 Terry and Kim each sat twenty class tests. Terry's results on the tests are displayed in the box-and-whisker plot shown in part (i).

- a. Kim's 5-number summary for the tests is 67, 69, 71, 73, 75. Draw a box-and-whisker plot to display Kim's results below that of Terry's results.

1



- b. What percentage of Terry's results were below 69?

50%

1